

Wearable Robotics

L10: Permanent Magnet Synchronous Motors (PMSM)

Prof. Edgar Bolívar

Sep. 25th, 2023

Permanent Magnet Synchronous Motor

Modeling three-phase brushless motors as a single-phase brushed motor

Field Oriented Control

Clarke-Park transform

Main components

Three-phase architecture

Electronic commutation

The issue of terminology and lack of standards

Orientation of the field
relative to a permanent magnet.

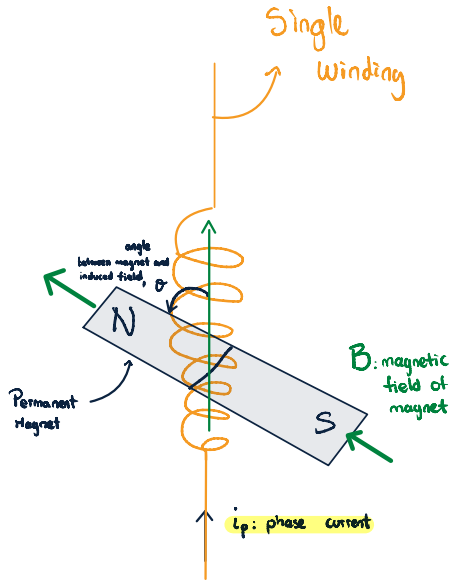
Field Oriented Control



Magnetic field
that you can induce
passing currents through
a wounded wire (winding)

↳ strategy to
select the winding
current

Let's analyze the behavior of a **single phase** and a **permanent magnet** that is constrained to translate but free to rotate



$$\tau = NAB \sin(\theta) \hat{i}_p$$

$$= k_{t-p} \hat{i}_p$$

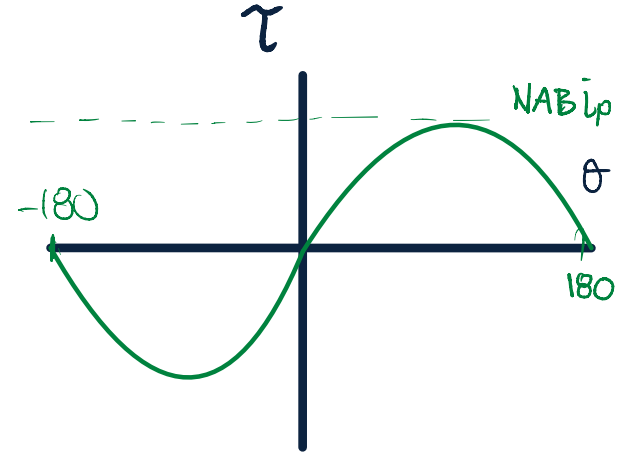
τ : torque between winding and magnet

N : winding turns

A : cross-section area of wire

$$\tau = k_{t-p} \hat{i}_p$$

Torque constant that depends on orientation



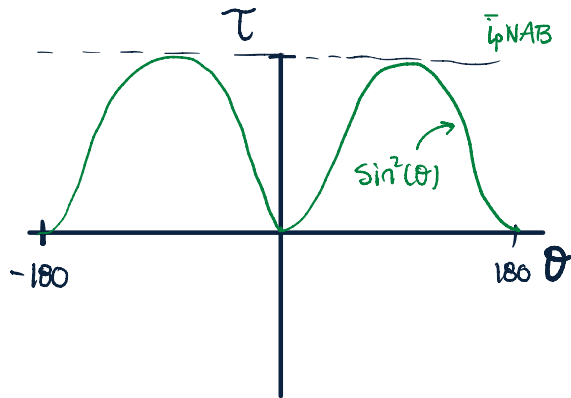
Q: How can we create a constant torque output?

A: Adding two more phases and using a sinusoidal torque profile

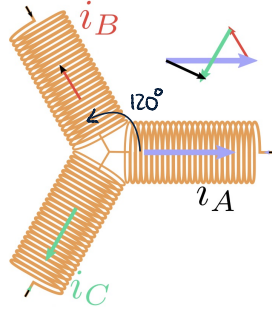
If we make the phase current

sinusoidal $\boxed{\bar{i}_p = \bar{i}_p \sin(\theta)}$ then

$$\tau = \bar{i}_p NAB \sin^2(\theta)$$



If we arrange three phases
120° apart



The total torque

$$\tau = \tau_A + \tau_B + \tau_C$$

$$= NBA \bar{i}_p [\sin^2(\theta) + \sin^2(\theta + 120^\circ) + \sin^2(\theta - 120^\circ)]$$

Great trigonometric news:

$$\sin^2(\theta) + \sin^2(\theta + 120^\circ) + \sin^2(\theta - 120^\circ)$$

$$= \frac{3}{2}$$



Thus

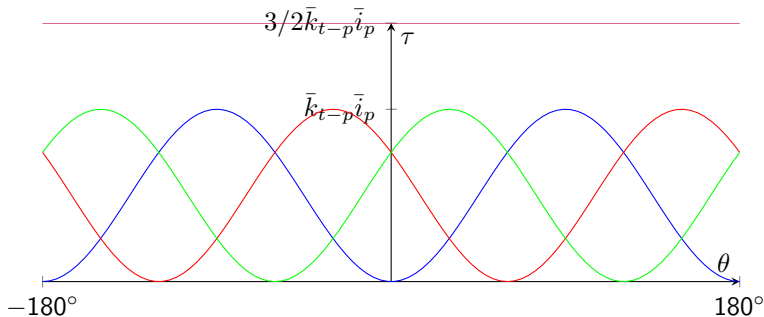
$$\boxed{\tau = \frac{3}{2} NBA \bar{i}_p}$$

Torque for three phases, 120° apart

$$* \bar{k}_{t-p} = NAB$$

$$\begin{aligned}\tau &= \bar{k}_{t-p} \bar{i}_p (\sin^2(\theta) + \sin^2(\theta + 120^\circ) + \sin^2(\theta + 240^\circ)) \\ &= \frac{3}{2} \bar{k}_{t-p} \bar{i}_p\end{aligned}$$

Three phases, 120° apart, generate a constant torque! (simplified math)



So far, we need to control three independent currents (i_a, i_b, i_c) to get a constant torque profile. How do we represent this three-phase motor as a single-phase motor?

$$\hat{i}_a = \bar{i}_a \sin(\theta)$$

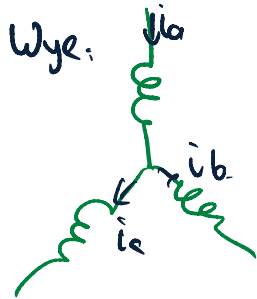
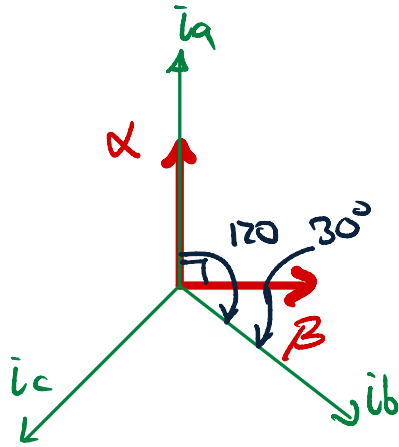
$$\hat{i}_b = \bar{i}_b \sin(\theta + 120)$$

$$\hat{i}_c = \bar{i}_c \sin(\theta + 240)$$

That's why we need the Clarke-Park transforms

Clarke transform:

Transform the **three** magnetic field/winding currents into **two** perpendicular axes (**α** and **β**)



$$\bar{i}_a = \bar{i}_b + \bar{i}_c$$

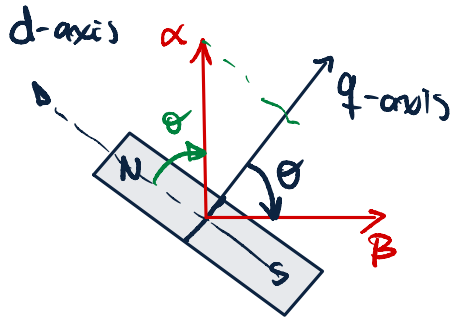
$$\bar{i}_\alpha = \bar{i}_a - \sin(30^\circ) \bar{i}_b - \sin(30^\circ) \bar{i}_c$$

$$\bar{i}_\beta = \cos(30^\circ) \bar{i}_b - \cos(30^\circ) \bar{i}_c$$

$$\begin{bmatrix} \bar{i}_\alpha \\ \bar{i}_\beta \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \bar{i}_a \\ \bar{i}_b \\ \bar{i}_c \end{bmatrix}$$

Clarke transform

Park transform: Transform the two magnetic fields/currents in the fixed frame into the moving frame of the magnet $\rightarrow$ (d-q axes)



$$i_d = -i_\alpha \cos(\theta) + i_\beta \sin(\theta)$$

$$i_q = +i_\alpha \sin(\theta) + i_\beta \cos(\theta)$$

d-axis: direct current. Current that induces a magnetic field in the direction of the permanent magnet. It does not produce torque

q-axis: quadrature current current that induces a magnetic field perpendicular to the field generated by the magnet.

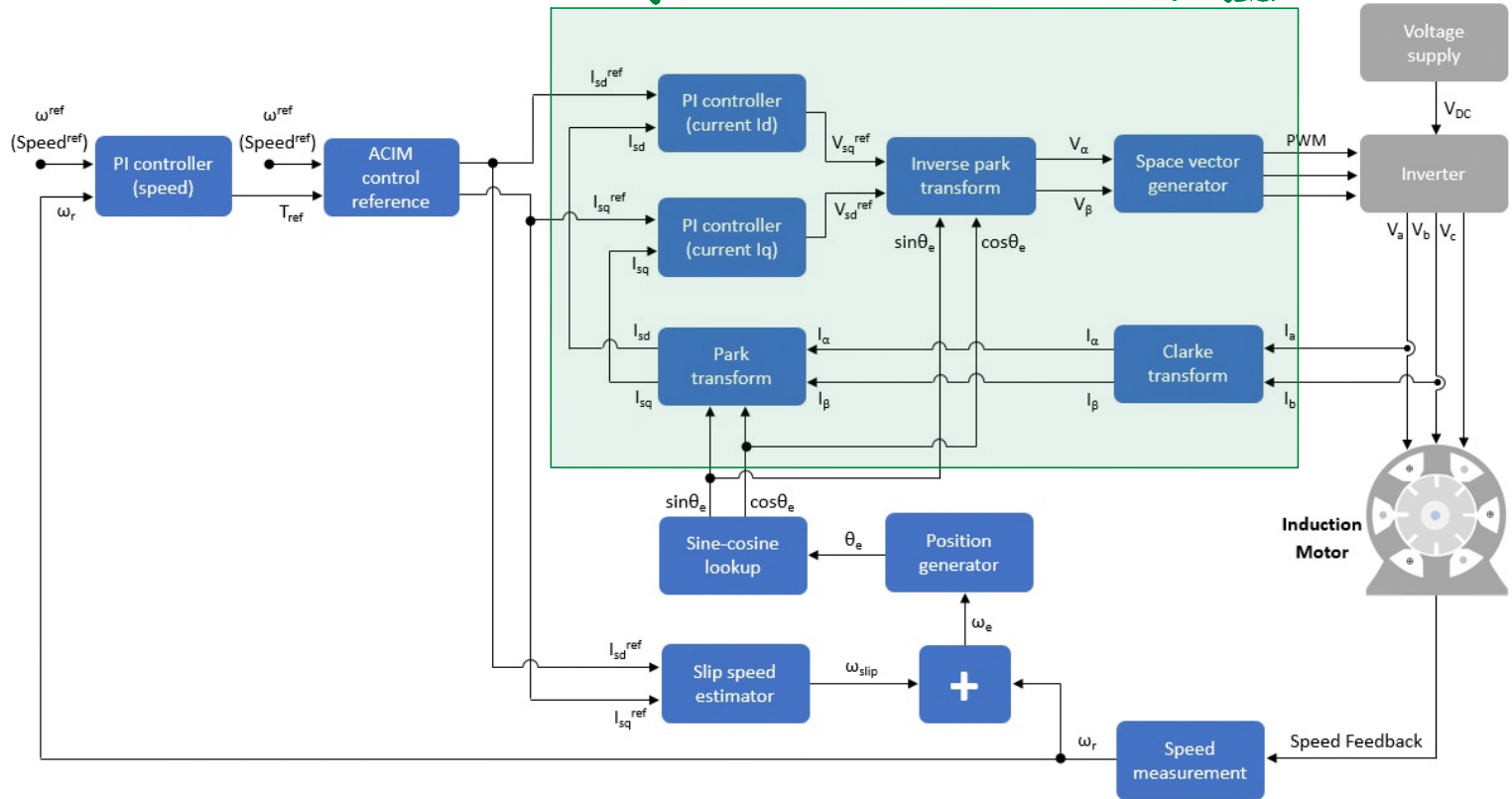
It generates torque

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}$$

↓
Park transform, it depends on the relative angle between the magnet and the stator.

Field Oriented Control

Main Strategy: Control the q-axis current to generate the desired torque.
Control the d-axis current to be zero.



* Image from MathWorks

Summary:

- 1) Transform the three winding currents/magnetic fields into the d-q axes currents

* d-axis current = 0

- 2) Control the q-axis current to generate the torque that you want $\tau_m = K_{Tq} i_q$

- 3) Expressing voltages and currents in the q-axis

We can model the brushless motor as a brushed motor

$$V_{in} = i_q R_q + L_q \frac{di_q}{dt} + V_{emf}$$

$$\begin{aligned} \tau_m &= I_m \ddot{\theta}_m + b_m \dot{\theta}_m + \mu \text{sign}(\dot{\theta}_m) - \tau_e \\ &= K_r i_q \end{aligned}$$

* System ID of PMSMs in next class
→ i_q, R_q, L_q, V_{emf} are in the q-axis.

Check Lee et al. for details.

Current in the winding $<$ Wye winding
Delta

Why multiple magnets in the rotor?

Kahoot!

Brushless DC motors are the
same as PMs.

Identification of Wye and Delta

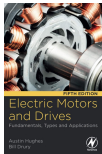
Notation

It has to be admitted that the industrial and academic communities have served to make life confusing in this area by giving an array of different names to essentially the same machine, so we begin by looking at the terminology. The names we will encounter include:

- (a) Conventional *Synchronous Machine* with its rotor field winding (excited-rotor). This is the only machine that may have brushes, but even then they will only carry the rotor excitation current, not the main a.c. power input.
- (b) *Permanent Magnet Synchronous Machine* with permanent magnets replacing the rotor field winding.
- (c) *Brushless Permanent Magnet Synchronous Motor* (same as (b)). The prefix 'brushless' is superfluous.
- (d) *Brushless a.c. Motor* (same as (b)).
- (e) *Brushless d.c. Motor* (same as (b) except for detailed differences in the field patterns). This name was coined in the 1970s to describe 'inside out' motors that were intended as direct replacements for conventional d.c. motors, and in this sense it has some justification.
- (f) *Permanent Magnet Servo Motor* (same as (b)).

Electric Motors and Drives, <https://doi.org/10.1016/B978-0-08-102615-1.00009-X>
© 2019 Elsevier Ltd. All rights reserved.

307



MATLAB / Simscape notation

BLDC v.s. PMSM both are brushless
but the back EMF profile changes.

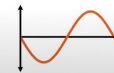
BLDC

Trapezoidal back-EMF
Controlled by trapezoidal control



PMSM

Sinusoidal back-EMF
Controlled by field-oriented control



Sometimes PMSM and BLDC motor terms
are used interchangeably.

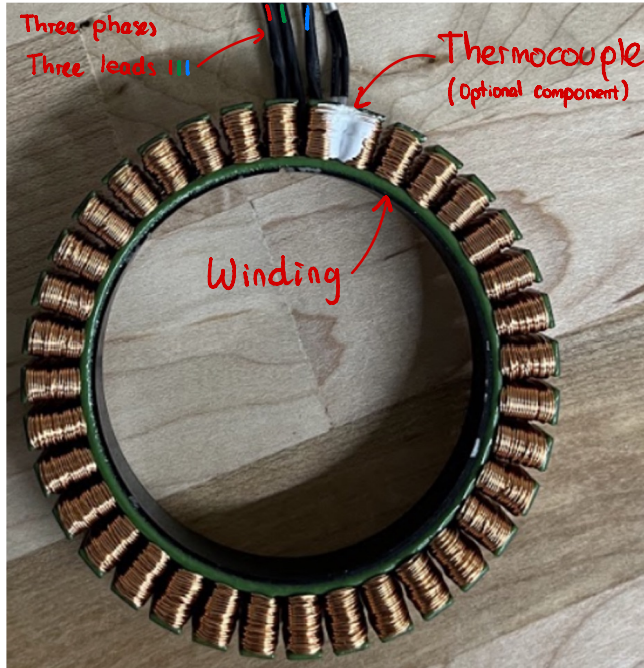
Motor Control, Part 1: An Introduction to Brushless DC Motors

MATLAB YouTube video

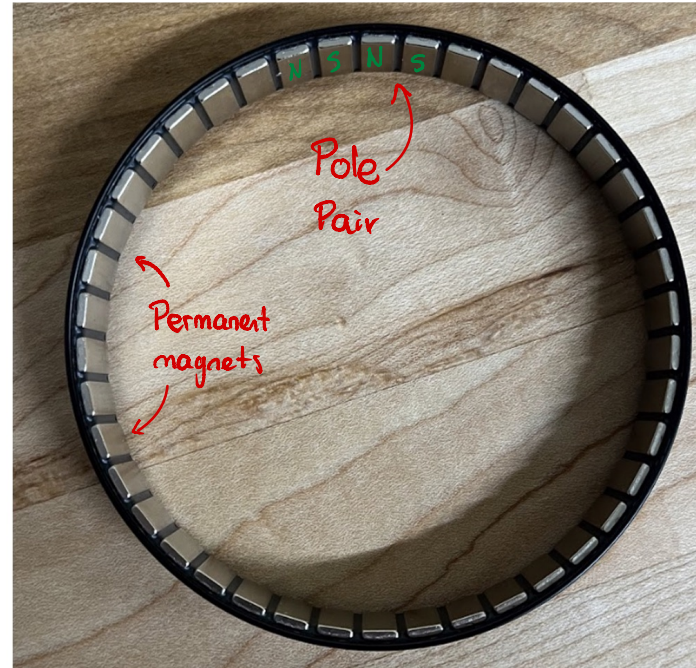
Permanent
Magnet
Synchronous
Motor/Machine

(PMSM)

Stator

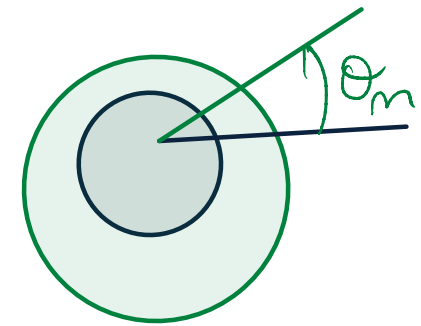


Rotor



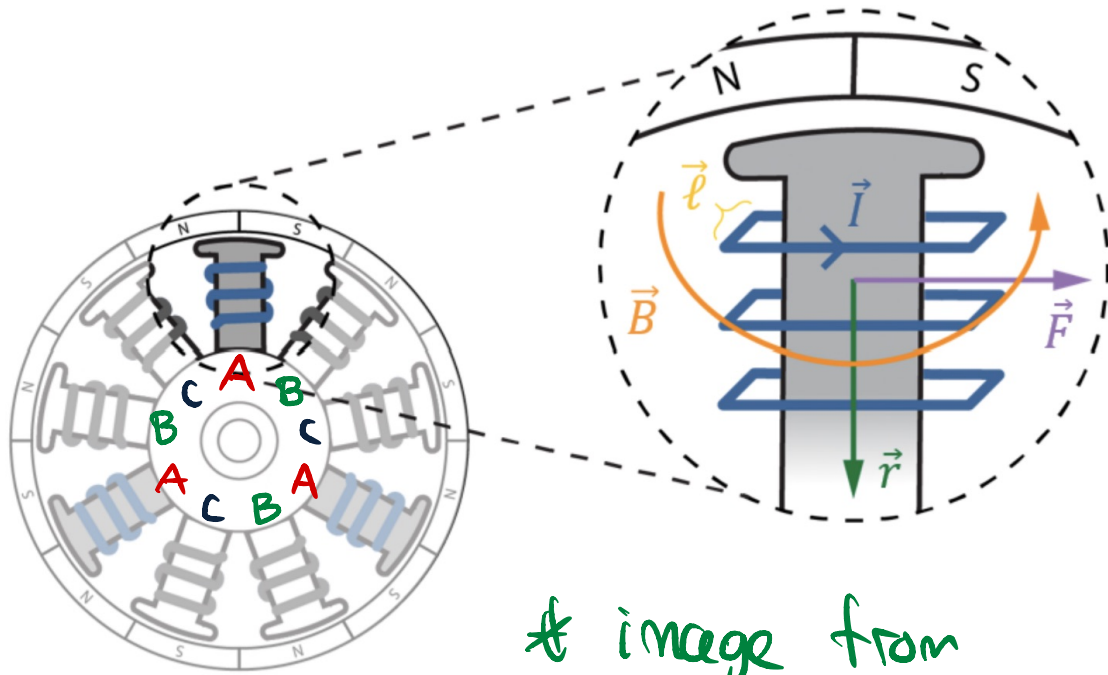
Wait, I see three wires but way more than three windings
I also see more than one magnet pair...

The mechanical angle.



The electrical angle.

$$\sigma_R = N \sigma_m$$



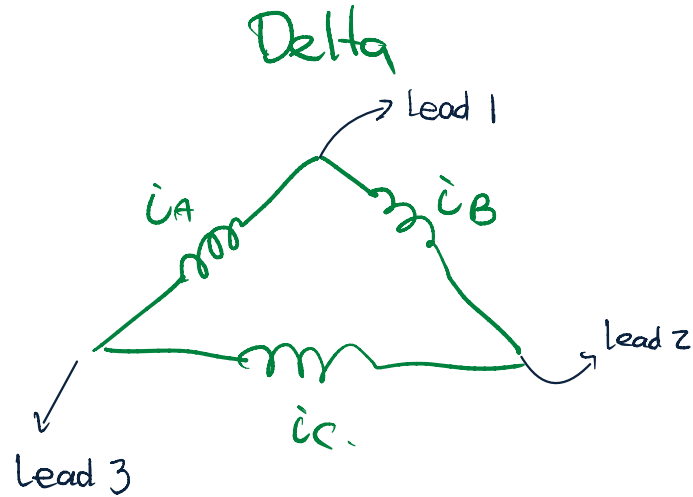
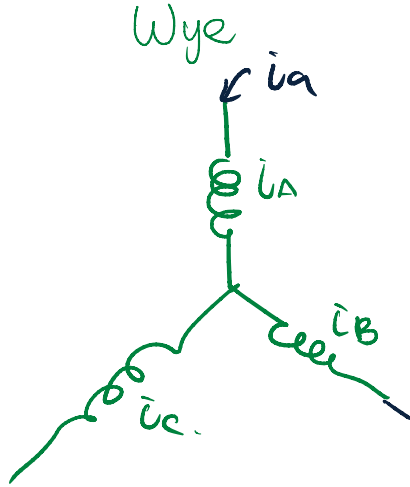
* image from
Lee et al.

6-pole pairs.

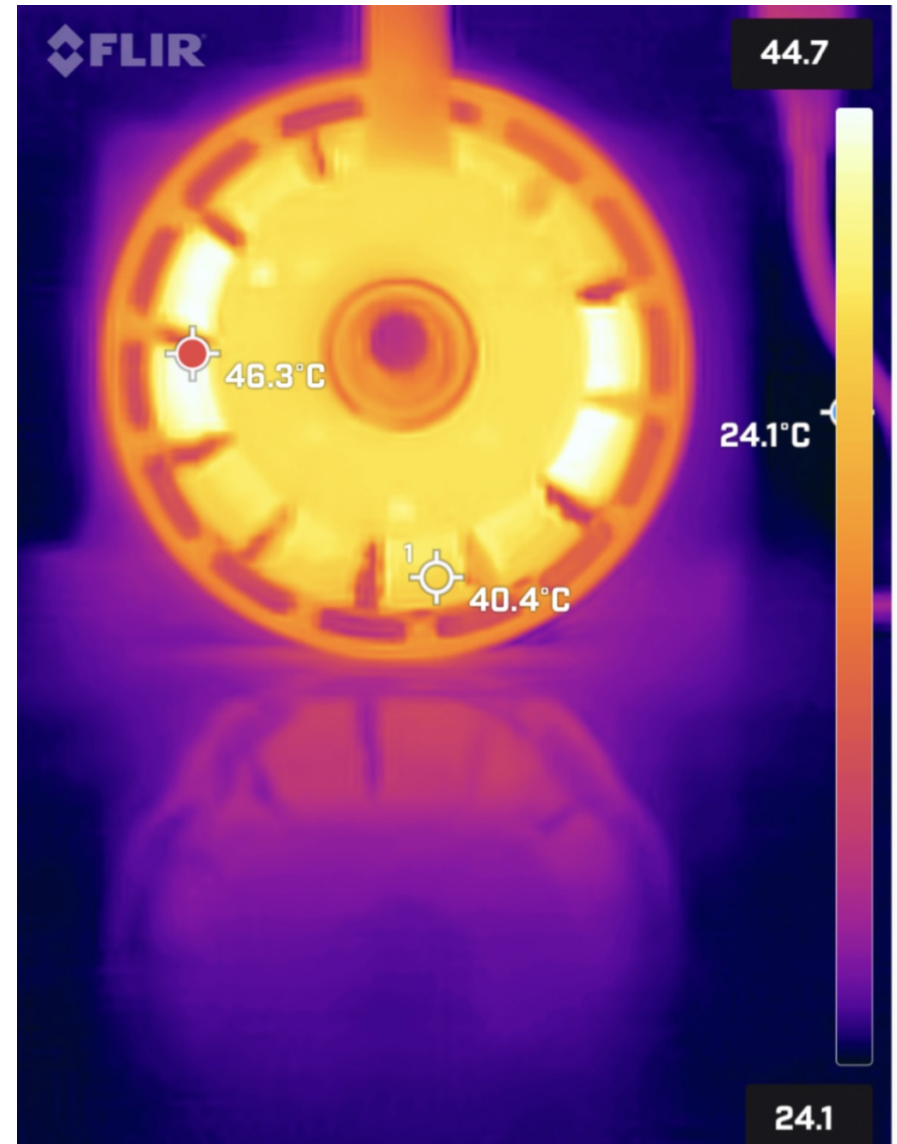
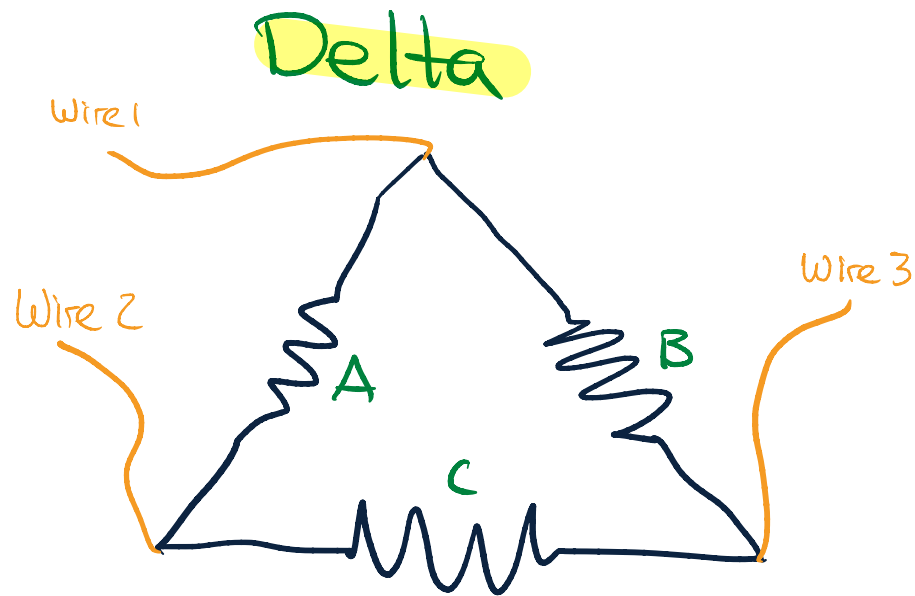
2

Three-phase architecture

The windings can be electrically arranged in multiple configurations.
two popular ones are



Thermal response from a delta configuration after applying a constant current through two of the leads.



Brushed motors:

Pros:

Simple to implement and model
(PWM signal + H-bridge)

Cons: (compared to brushed)

- Greater friction from brushes
- Lower speeds
- Lower specific power (W/kg)

Brushless motors:

Pros:

- Less friction
- High speed and specific power

Cons:

- Requires electrical commutation
(High computing power is required)

* TQ- RoboDrive datasheet.